ELEC70054
ELEC60022

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

MICROWAVE TECHNOLOGY

Wednesday, 14 December 10:00 am

Time allowed: 3:00 hours

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : O. Sydoruk

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Special instructions for invigilators

This is an OPEN BOOK examination with the students only allowed to bring-in an unaltered (i.e. without annotations of any sort) set of officially issued lecture PowerPoint slides.

Tutorial question solutions (i.e. from past exam papers) are not permitted.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSARY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3 (three) questions, each carrying a weighting of 25%.

All variables have their usual meaning.

Permeability of free space, $\boldsymbol{\mu_o = 4\pi \times 10^{-7}}$ $\mathbf{[H/m]}$

Permittivity of free space, $\boldsymbol{\varepsilon_o \approx 8.854}$ $\mathbf{[pF/m]}$

1.

Consider a parallel-plate transmission line (operating in the dominant TEM mode) with a width W = 1 mm and separation distance of h = 0.01 mm. Assume both plates are made from T = 0.01 mm thick copper, having a conductivity of 5.8 x 10^7 S/m. The line is embedded with a homogenous medium of PTFE, having a dielectric constant of 2.1, loss tangent of 2.1 x 10^{-4} and dielectric strength (breakdown field strength) of 10 kV/mm.

Calculate the following at 2 GHz and include the units:

i) Total DC resistance in Ω/m. [2]

ii) Surface resistance and plate thickness to skin depth ratio. [4]

iii) Distributed resistance. [2]

iv) Distributed inductance. [2]

v) Distributed conductance. [4]

vi) Distributed capacitance. [2]

vii) Lossless characteristic impedance. [2]

viii) Lossy characteristic impedance. [3]

ix) Lossless phase velocity. [2]

x) Lossless phase constant. [2]

xi) Attenuation constant due to the conductors only. [2]

xii) Attenuation constant due to the dielectric only. [2]

xiii) Total attenuation constant and insertion loss in dB/m. [4]

xiv) Q-factor associated with the conductors only. [2]

xv) Q-factor associated with the dielectric only. [2]

xvi) Overall Q-factor. [2]

xvii) Maximum peak voltage. [2]

xviii) Maximum peak power under impedance matched conditions. [2]

xix) Lowest-order transverse resonant mode frequency. [4]

xx) Maximum frequency for TEM mode of operation. [3]

2. This question relates to the classical skin effect of normal metals.

a) State the main application(s) in the microwave industry where aluminium is used at room temperature and at liquid nitrogen temperatures. Briefly explain the reasons for your answers.

[3]

b) Show how the classical skin depth can be calculated from the propagation constant.

[2]

c) Show how the complex skin depth can be calculated from the propagation constant and express the real and imaginary parts in terms of the classical skin depth.

[2]

d) Derive the simple expressions for both the surface resistance and surface inductance, in terms of the classical skin depth.

[4]

e) Aluminium has a bulk DC conductivity of $\sigma_o = 3.8 \times 10^7$ S/m at room temperature. For an operating frequency of 1 GHz, calculate the following:

i) Classical skin depth.

[2]

ii) Wavelength inside the metal.

[2]

iii) Surface impedance.

[2]

iv) Surface resistance.

[1]

v) Surface inductance.

[2]

f) Draw the telegraphers equation's one-dimensional equivalent circuit models for propagation of a TEM wave both in free space at normal incidence to a metal boundary and also within the metal. Clearly label all the distributed lumped elements and give their units.

[5]

3. An ideal air-filled rectangular waveguide has a guided-wavelength given by the following expression:

$$\lambda_g = \frac{\lambda_o}{\sqrt{1-\left(\frac{f_c}{f_o}\right)^2}} \tag{3.1}$$

All variables have their usual meaning.

In addition, for the TE_{101} mode, the unloaded Q-factor for an air-filled rectangular waveguide resonant cavity is given by the following expression:

$$Q_u|_{TE101} \cong \frac{Volume}{Area \times(\delta_o\ /2)} \tag{3.2}$$

All variables have their usual meaning.

a) For a half-height waveguide: height dimension *b* is half that of the width dimension *a*:

i) Using (3.1), derive an expression for the length *l* of the cavity in terms of *a* and the various frequency terms.
[5]

ii) Using (i), derive an expression for the internal volume of the cavity.
[3]

iii) Using (i), derive an expression for the internal area of the cavity.
[4]

iv) Using (3.2) and assuming that $f_o/f_c = \sqrt{2}$, derive an expression for the unloaded-Q-factor in terms of *a* and classical skin depth.
[4]

v) Using (iv), calculate the unloaded-Q-factor for a 15.5 GHz resonant cavity made with copper walls having a bulk DC conductivity of $\sigma_o = 5.8 \times 10^7$ S/m.
[5]

b) For a cubical cavity (i.e. all internal dimensions are equal) derive an expression for the unloaded-Q-factor in terms of *a* and classical skin depth and show that this has a 33.333% higher unloaded Q-factor.
[4]

4.

a) Briefly describe the various steps needed to derive a transfer function using the Mason's non-touching loop rule and state the general form of the corresponding equation for a transfer function.

[5]

b) Two amplifiers are connected in cascade, with the final output stage terminated with a load impedance having an arbitrary value. The amplifiers can be defined by the scattering (S)-parameter matrices $[S_a]$ and $[S_b]$, and the load impedance can be expressed by its voltage-wave reflection coefficient ρ_L. Draw the corresponding signal flow graph and determine the exact expression for the overall input voltage-wave reflection coefficient ρ_{IN}.

[14]

c) For simplicity, phase angle is ignored and the amplifiers in 4(b) are assumed to be identical. Given the following:

$$[S_a] = [S_b] = \begin{bmatrix} 0.1 & 0.1 \\ 10 & 0.1 \end{bmatrix} \quad \text{and} \quad \rho_L = 0.5 \tag{4.1}$$

Calculate the overall input return loss for the circuit in 4(b) and comment on how this has changed by introducing the amplifiers.

[6]

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